INCH-POUND

MIL-D-35033E
21 November 1989
SUPERSEDING
MIL-D-35033D
6 December 1977

MILITARY SPECIFICATION

DESSERT POWDER, PUDDING, INSTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers flavored instant pudding desserts, prepared by mixing the powder with reconstituted nonfat dry milk. The product is intended for use by the Department of Defense for operational rations and as an item of general issue.

1.2 Classification. The dessert powder shall be of the following flavors as specified (see 6.1):

Chocolate
Vanilla
Butterscotch

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

PPP-C-29 - Canned Subsistence Items, Packaging Of
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1120, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Bid sample approval. Unless otherwise specified (see 6.1), duplicate individual samples of at least 2 pounds of each flavor of product that the contractor proposes to furnish, packaged in hermetically sealed cans (or other package which will preclude deterioration of the product), shall be submitted to the contracting officer prior to bid opening. One sample of each duplicate shall be evaluated to determine compliance with 3.4 for the subjective characteristics (flavor, odor, and texture). Color requirements shall be in compliance with 3.4.1. In addition, the samples shall be rated as to overall quality, taking into consideration all the subjective characteristics. The duplicate approved bid sample shall be used as a standard for the subjective characteristics in determining acceptance of intended delivery product. The approval of any bid sample for the aforementioned characteristics will not constitute approval of the product as meeting the other requirements of this document.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Sugar. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof, and shall be "Bakers Special" granulation.

3.2.2 Starch. Starch shall be derived from either waxy maize, dent corn, tapioca, or a combination thereof. The starch shall be modified, cold water swellable, and shall have the following typical analyses:

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Chemical
pH: 4.5 - 7.0
Moisture: 9 percent maximum

Particle size
thru U.S. Standard #100 sieve - 95 percent

Viscosity (see 6.4) - Brabender-700 cmg cartridge (head) 1/
Peak viscosity 875-1425 Brabender Units (B.U.)
Percent viscosity drop 58-77 percent of peak maximum

1/ If 350 Head is used, multiply Brabender Units by 2.

3.2.3 Cocoa. Cocoa powder shall be prepared from nibs of domestically roasted, mature, well fermented, sound and wholesome cocoa beans, which have been properly dried, cured, and mildly alkalized in accordance with the definitions and standards of the Food and Drug Administration. Neither sodium carbonate nor sodium bicarbonate shall be used in dutching (alkali treatment) process. The pH shall be not less than 6.0 nor more than 7.5, and the fat content (cocoa butter) shall be not less than 10 percent. Chemically extracted cocoa, in part or whole, shall not be acceptable. When washed with petroleum ether, not less than 98 percent by weight shall pass through a U.S. Standard No. 200 sieve. No expeller process cocoa shall be acceptable.

3.2.4 Powdered shortening. Powdered shortening shall be spray dried, free flowing, and free of burnt or scorched aroma. The powdered shortening shall be composed of partially hydrogenated vegetable oil, sodium caseinate, corn syrup solids, soy protein isolate, dipotassium phosphate, sodium silicoaluminate, tricalcium phosphate with or without antioxidant and shall have the following typical analyses:

Chemical analyses

Total oil, percent	75 ± 2.5
Protein, percent	5.5 ± 0.5
pH	7.5 ± 0.5
Moisture, percent	1.5 ± 0.5
Carbohydrate, percent	16.0 ± 1.0
Ash, percent	2.0 ± 1.0

3.2.5 Dextrose. Dextrose shall be free flowing dextrose hydrate powder and shall comply with the Food Chemicals Codex.

3.2.6 Butterscotch flavor. Butterscotch flavor shall be a natural or synthetic spray dried flavor.

3.2.7 Tetrasodium pyrophoshate, anhydrous. Anhydrous tetrasodium pyrophosphate shall comply with the requirements of the Food Chemicals Codex, shall be water soluble, and shall be pudding grade.

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3.2.8 Calcium acetate. Calcium acetate shall comply with the requirements of the Food Chemicals Codex.

3.2.9 Salt. Salt shall be noniodized, white, refined sodium chloride, with or without anticaking agents, and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.10 Disodium phosphate, anhydrous. Anhydrous disodium phosphate shall comply with the requirements of the Food Chemicals Codex.

3.2.11 Sodium caseinate. Sodium caseinate shall be spray dried, food grade sodium caseinate made from fresh skim milk. The sodium caseinate shall be white or a very light cream color.

3.2.12 Titanium dioxide. Titanium dioxide shall comply with the requirements of the Food Chemicals Codex, and shall be water dispersible.

3.2.13 Vanilla flavor. Vanilla flavor shall be pure vanilla or artificial vanilla flavor or a combination thereof.

3.2.14 Yellow No. 5, Yellow No. 6, and butterscotch shade. All colors shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. The butterscotch shade shall be a monoblend spray-dried mixture of the following colors.

Color	Percent by weight
FD&C Yellow No. 5	75
FD&C Red No. 40	23
FD&C Blue No. 1	2

In addition, the coloring material shall be ground sufficiently so that it shall readily dissolve without forming color specks (see 3.4).

3.3 Formulation. The instant dessert powder shall be prepared according to the formulation below.

Ingredients	Parts by weight		
	Chocolate	Vanilla	Butterscotch
Sugar	59.0 to 63.0	69.0 to 72.0	69.0 to 72.0
Starch	15.0	16.0 to 17.0	16.0 to 17.0
Cocoa	13.0 to 13.5	--	--
Powdered shortening	5.0 to 5.5	5.0 to 7.5	5.0 to 7.5

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Ingredients	Parts by weight		
	Chocolate	Vanilla	Butterscotch
Dextrose	3.4 to 7.4	3.0 to 6.0	3.0 to 6.0
Butterscotch flavor	--	--	1.5 to 2.0
Tetrasodium pyrophoshate, anhydrous	2.35	1.2	1.3
Calcium acetate	1.2	--	--
Salt	0.6	1.1	1.1
Disodium phosphate, anhydrous	--	0.9	0.9
Sodium caseinate	0.50 to 0.55	0.50 to 0.55	0.50 to 0.55
Titanium dioxide	--	0.20	0.40
Vanilla flavor	--	0.15 to 0.20	--
Butterscotch shade	--	--	0.05 to 0.07
FD&C Yellow No. 5	--	0.007 to 0.01	--
FD&C Yellow No. 6	--	0.002 to 0.0025	--

3.4 Finished product requirements. The finished product shall comply with the following requirements:

Dessert powder:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

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d. The product shall be a well-mixed, free flowing powder of such particle size that not less than 99 percent shall pass through a U.S. Standard No. 35 sieve.

e. The moisture content shall be not more than 3.5 percent, by weight.

Prepared product (prepared as specified in footnote 3 of table II):

f. Prepared pudding and pie filling shall be creamy, smooth textured, devoid of lumps, color specks, and graininess.

g. Prepared pudding and pie filling shall have the flavor, odor, and appearance characteristic of the specified flavor of pudding and pie filling, as applicable.

h. Prepared pie filling shall have sufficient rigidity without being rubbery or gummy and shall have clean cut edges when sliced.

i. Sliced portions of prepared pie filling shall show no evidence of weeping for at least three hours after preparation.

3.4.1 Color. The prepared product shall meet the following color requirements.

	Color range
Flavor 1 (Chocolate)	Not lighter than 20109 nor darker than 20061 of FED-STD-595
Flavor 2 (Vanilla)	Not lighter than 23793 nor darker than 23695 of FED-STD-595
Flavor 3 (Butterscotch)	Not ligher than 10371 nor darker than 30257 of FED-STD-595

3.4.2 Palatability. The finished product shall be equal to or better than the approved bid sample in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing, or Holding of Human Food", and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to ensure compliance with the requirements of this specification.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, US Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.3.2 In-process examination. In-process examination shall be performed to determine conformance to the formulation, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of formulation or of other valid documents shall be cause for rejection of the involved product.

4.3.3 Net weight examination. The filled and sealed cans shall be examined for net weight. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. Each flavor shall be examined as a separate lot. The inspection level shall be S-3. Results shall be reported to the nearest 1 ounce. Acceptance criteria shall be as specified in the procurement documents (see 6.3).

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TABLE I. Net weight defects 1/

Category	Defect
Minor	
201	Net weight of individual can less than 5 pounds 7 ounces for vanilla or butterscotch flavored product
202	Net weight of individual can less than 4 pounds 15 ounces for chocolate flavored product

1/ The lot shall be rejected if the sample average net weight is less than 5 pounds 8 ounces for vanilla or butterscotch flavored product or is less than 5 pounds for chocolate flavored product.

4.3.4 Finished product examination. The finished product shall be examined for the defects listed in tabie II. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. Each flavor shall be examined as a separate lot. The inspection level shall be S-2. Acceptance criteria shall be as specified in the procurement documents (see 6.3).

TABLE II. Product defects 1/ 2/

Category		Defect
Major	Minor	
		Dry dessert powder
101		Not free flowing
102		Not uniformly mixed
		Prepared product 3/
103		Not creamy or smooth textured, exhibits rubbery or gummy texture, or presence of lumps, color specks, or graininess.
104		Product does not have flavor, odor, and appearance characteristic of the specified flavor of pudding or pie filling

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TABLE II. Product defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
		Prepared product 3/ (cont'd)
105		Sliced portion of pie shows evidence of weeping within three hours after preparation
	201	Presence of unclean cut edge of sliced portion of pie
	202	Color not as specified

1/ The presence of foreign material (for example, glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved bid sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service agent.)

3/ The dessert powder shall be prepared as pudding or pie filling as described below. Unless otherwise specified, the mix shall be prepared as pie filling and shall be filled into a 9 inch pan with graham cracker or baked pie shell.

For pie filling: Seven ounces of dessert powder shall be added to 20 fluid ounces of reconstituted nonfat dry milk and mixed for 2 to 3 minutes (15 seconds at low speed and 2 minutes at medium speed). Chill at least 1 hour.

For pudding: Seven ounces of dessert powder shall be added to 24 fluid ounces of reconstituted nonfat dry milk and mixed for 2 to 3 minutes (15 seconds at low speed and 2 minutes at medium speed). Chill at least 1 hour.

4.3.5 Examination for particle size of finished product. The dessert powder shall be examined for particle size. Place a 100-gram composite sample on the specified sieve (see 3.4) with pan attached. Set on a flat, smooth, horizontal surface. Using a soft 3-inch hair brush, with handle pointing towards the operator at an angle of about 45 degrees, stroke the sample along the perimeter of the sieve by drawing the brush towards the operator in 5-to-7-inch strokes. The full width of the brush should bear firmly upon the mesh and follow the curvature of the perimeter. Rotate the sieve about 1/16 turn after each stroke in a direction opposite the motion of the brush. Brush at the rate of one stroke per second for 5 minutes, or until the end point has been reached.

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Report the percent materials passing through the sieve. The sample for examination shall be a 1 pound composite of each flavor derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of cans. One determination shall be made utilizing 100 grams of the product from the composite. Results shall be reported to the nearest percent. Any result failing to conform to the particle size requirement in 3.4 shall be classified as a major defect and shall be cause for rejection of the lot.

4.3.6 Moisture content determination. The dessert powder shall be tested for moisture content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; chapter Sugars and Sugar Products; section Sugars, Sirups; Moisture - Vacuum Drying, Official Final Action Method. The sample for testing shall be an 8 ounce composite of each flavor derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of cans. Results shall be reported to the nearest 0.1 percent. Any result failing to conform to the moisture requirement in 3.4 shall be classified as a major defect and shall be cause for rejection of the lot.

4.3.7 Examination of can. Examination of the cans shall be in accordance with examination criteria of PPP-C-29.

4.3.8 Examination of can labeling. Examination of can labeling shall be in accordance with the examination criteria of MIL-L-1497.

4.3.9 Shipping container examination. Shipping containers shall be examined in accordance with applicable quality assurance provisions of PPP-C-29.

4.3.10 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C, as specified (see 6.1).

5.1.1 Level A or C (cans). Five pounds of chocolate flavored dessert powder or 5 pounds 8 ounces of vanilla or butterscotch flavored dessert powder shall be unit packed in a size 603 by 700 can in accordance with the level A or C requirements of PPP-C-29.

5.2 Packing. The product shall be packed in accordance with levels A, B, or C, as specified (see 6.1).

5.2.1 Levels A and B packing. Cans shall be packed in accordance with the level A or B requirements of PPP-C-29, as specified (see 6.1).

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5.2.2 Level C packing. The shipping container shall be in accordance with the National Motor Freight Classification or Uniform Freight Classification, as applicable, except closure and reinforcement of the fiberboard boxes shall be in accordance with method II as specified in the appendix of PPP-B-636.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2.1 or 5.2.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling and marking.

5.4.1 Cans. Cans shall be labeled in accordance with MIL-L-1497. In addition, the following directions for use shall be lithographed on the body or on one end of the can.

DIRECTIONS FOR USE

CHOCOLATE FLAVOR

(1) Chocolate pudding: Place 8 1/2 quarts chilled, reconstituted nonfat dry milk in a mixer bowl. Add contents of No. 10 can dessert powder. Using whip, blend at low speed 15 seconds or until blended. Scrape sides and bottom of bowl; whip at medium speed 2 minutes or until smooth. Chill at least 1 hour.

YIELD: 10 quarts or 80 (1/2 cup) portions.

THIS IS A COLD PREPARATION - DO NOT HEAT

(1 pound, 12 ounces (6 1/2 cups) nonfat dry milk, combined with 8 quarts cold water will yield 8 1/2 quarts reconstituted, chilled, nonfat milk.)

(2) Chocolate pie filling: Place 7 quarts chilled, reconstituted nonfat dry milk in a mixer bowl. Add contents of No. 10 can dessert powder. Using whip, blend at low speed 15 seconds or until blended. Scrape sides and bottom of bowl; whip at medium speed 2 minutes or until smooth. Pour about 3 1/8 cups filling into each of eleven baked 9-inch pie shells. Chill at least 1 hour. Cut eight wedges per pie.

YIELD: 8 3/4 quarts pie filling (88 portions).

THIS IS A COLD PREPARATION - DO NOT HEAT

(1 pound, 7 ounces (5 1/4 cups) nonfat dry milk combined with 6 1/2 quarts cold water will yield 7 quarts reconstituted nonfat milk.)

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DIRECTIONS FOR USE

VANILLA OR BUTTERSCOTCH FLAVOR

(1) Vanilla or butterscotch pudding (as applicable): Place 9 1/4 quarts chilled, reconstituted nonfat dry milk in a mixer bowl. Add contents of No. 10 can dessert powder. Using whip, blend at low speed 15 seconds or until blended. Scrape sides and bottom of bowl; whip at medium speed 2 minutes or until smooth. Chill at least 1 hour.

YIELD: 11 quarts or 88 (1/2 cup) portions.

THIS IS A COLD PREPARATION - DO NOT HEAT

(One pound, 14 ounces (7 cups) nonfat dry milk combined with 8 3/4 quarts cold water will yield 9 1/4 quarts reconstituted nonfat milk.)

(2) Vanilla or butterscotch pie filling (as applicable): Place 7 3/4 quarts chilled, reconstituted nonfat dry milk in a mixer bowl. Add contents of No. 10 can dessert powder. Using whip, blend at low speed 15 seconds or until blended. Scrape sides and bottom of bowl; whip at medium speed 2 minutes or until smooth. Pour about 3 cups filling into each of thirteen baked 9-inch pie shells. Chill at least 1 hour. Cut eight wedges per pie.

YIELD: 9 3/4 quarts pie filling (104 portions).

THIS IS A COLD PREPARATION - DO NOT HEAT

(One pound, 9 ounces (5 3/4 cups) nonfat dry milk combined with 7 1/4 quarts cold water will yield 7 3/4 quarts reconstituted nonfat milk.)

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Marking of unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Flavor required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When other than duplicate 2-pound bid samples are to be furnished (see 3.1).
e. Level of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load when unit loading is required (see 5.3).
g. Acceptance criteria required (see 6.3).

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Acceptance critieria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise indicated, the following acceptance criteria are in accordance with MIL-STD-105.

Paragraph title	Recommended acceptable quality level (AQL)	
	Major	Minor
a. Net weight examination	---	2.5
b. Finished product examination	2.5	4.0

6.4 Viscosity determination of starch (CMLB-113T Brabender). The starch viscosity should be determined as follows (see 3.2.2):

a. A dry mix containing 26.55 grams (anhydrous weight basis) starch and 132.75 grams sugar (sucrose) is prepared. 1/

b. The required amount of distilled water and 10.8 grams glacial acetic acid are weighed into a mixing bowl (total weight of ingredients is 576 grams (g)). 2/

c. Over a period of 3 minutes, slowly add dry blend to bowl containing water and acid. During addition, the contents of the bowl are mixed using an electric mixer set at low mixing speed (approximately 75 rpm).

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d. Immediately scrape bowl contents into Brabender cup using a rubber spatula.

e. Turn Brabender instrument on with thermoregulator set at 30°C and hold at that setting for 10 minutes. It may be necessary to add Brabender weights to prevent chart pen from going off scale.

f. Set to controlled heating (1 1/2°C per minute increase) and heat to 95°C.

g. Hold 10 minutes at 95°C and take viscosity reading from chart.

1/ Amount of starch (anhydrous weight basis) used is determined as shown in following example. A moisture determination (infrared moisture balance) is done on the starch, which shows it to have 7 percent moisture. Weight of starch is:

$$\frac{26.55}{1.00 - 0.07} = 28.55 \text{ grams}$$

2/ Amount of distilled water used is determined as shown in following example:

Starch	28.55 grams
Sugar	132.75 grams
Acetic acid	10.80 grams
	172.10 grams

Total grams of ingredients used 576.0 grams.

576.0 grams - 172.1 grams = 403.9 grams water required.

6.5 Subject term (key word) listing.

Pie filling

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - SA
Air Force - 50

Preparing activity:
Army - GL

(Project 8940-0641)

Review activities:
Army - MD, TS
Navy - MC
DP - SS

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